CCE-321 Computer Peripheral and Interfacing

Final Examination, B.Sc. Engineering in CSE, Level III, Semester II (Session 2021-2022) | July-December 2024 | Dept. of CCE, PSTU, Dumki, Patuakhali | Time: 3 Hr | Marks: 70

Question and Model Answer (English + বাংলা ব্যাখ্যা)

Note / নোট: Answer any 5 out of 6 questions (5টি করতে হবে, তবে সব ৬টির উত্তর দেওয়া আছে)। Q1 ও Q2 স্লাইড-ভিত্তিক, তাই নিজের স্লাইডের শব্দের সাথে মিলিয়ে নিও।

Question 1 (2 × 7 = 14)

[A] i. What is the Internet of Things (IoT)? Explain its goal and how it differs from traditional computer networks.

IoT is a network of physical objects ("things") embedded with sensors, software and connectivity that collect and exchange data over the Internet with little or no human intervention.

Goal: to connect everyday objects so they can sense, communicate and act, making processes automated, efficient and data-driven.

Traditional computer network	IoT
Connects computers, phones, servers	Connects billions of sensors, actuators, machines
Human-generated data	Machine-generated data
Resource-rich devices	Constrained devices (low power, low memory)
Standard TCP/IP, high bandwidth	Lightweight protocols (CoAP, MQTT), low data rate

বাংলা: IoT হলো সেন্সর ও ইন্টারনেট যুক্ত বস্তুর নেটওয়ার্ক, যারা মানুষের হস্তক্ষেপ ছাড়াই ডেটা আদান-প্রদান করে। সাধারণ নেটওয়ার্কে কম্পিউটার/ফোন থাকে, কিন্তু IoT-তে ছোট, কম-শক্তির সেন্সর/ডিভাইস থাকে।

ii. List and briefly explain any three major benefits of IoT as mentioned in the slides.

1. **Efficiency and automation:** machines monitor and control themselves, so less manual work and less downtime.
2. **Cost saving:** predictive maintenance and energy optimisation reduce operating cost.
3. **Better decision making:** real-time data and analytics give accurate insights.

বাংলা: (১) কাজ স্বয়ংক্রিয় হয়, (২) খরচ কমে, (৩) রিয়েল-টাইম ডেটা থেকে ভালো সিদ্ধান্ত নেওয়া যায়।

iii. What is an IoT gateway? Describe its role in providing bidirectional communication and security.

A gateway sits between IoT end devices (sensors) and the cloud/Internet. It aggregates data and translates protocols (e.g. Zigbee/BLE to Ethernet/Wi-Fi/LTE).

- **Bidirectional communication:** upstream it sends sensor data to the cloud; downstream it delivers commands and firmware updates from the cloud to the devices.
- **Security:** device authentication, data encryption (TLS), firewall/filtering, secure boot and access control, so weak devices are not exposed directly to the Internet.

বাংলা: গেটওয়ে হলো সেন্সর ও ক্লাউডের মাঝের সেতু। এটি প্রোটোকল রূপান্তর করে, ডেটা উপরে (cloud) ও কমান্ড নিচে (device) পাঠায় এবং এনক্রিপশন/অথেনটিকেশন দিয়ে নিরাপত্তা দেয়।

iv. Name any three real-world IoT applications. Briefly explain how IoT is used in each.

- **Smart home:** smart thermostats, lights and locks are controlled automatically by sensors and phone apps.
- **Healthcare:** wearables monitor heart rate, SpO_2 and glucose, and alert doctors in an emergency.
- **Agriculture:** soil-moisture and weather sensors control irrigation automatically, saving water and raising yield.

বাংলা: স্মার্ট হোম (অটো লাইট/তাপমাত্রা), স্বাস্থ্য (ওয়্যারেবল দিয়ে রোগী মনিটর), কৃষি (মাটির আর্দ্রতা দেখে সেচ)।

v. Draw or describe the commonly used 4-layer IoT architecture. Explain the function of each layer.

```
+-------------------------------------+
| 4. Application Layer                |  smart home, city, health apps
+-------------------------------------+
| 3. Data Processing / Support Layer  |  storage, analytics, cloud
+-------------------------------------+
| 2. Network (Transport) Layer        |  gateways, Wi-Fi, LTE, LoRa
+-------------------------------------+
| 1. Perception / Sensing Layer       |  sensors, actuators, RFID
+-------------------------------------+
```

- **Perception layer:** senses/collects data from the physical world and performs actuation.
- **Network layer:** transmits data securely from devices to the processing system.
- **Data processing layer:** stores, filters and analyses data (cloud/edge) and makes decisions.
- **Application layer:** gives the end user services and an interface (dashboard, mobile app).

বাংলা: নিচ থেকে উপরে: Perception (ডেটা সংগ্রহ) → Network (ডেটা পাঠানো) → Data Processing (বিশ্লেষণ/সংরক্ষণ) → Application (ব্যবহারকারী সেবা)।

vi. Differentiate between licensed and unlicensed frequency bands in IoT. Give one example of each.

Licensed	Unlicensed
Needs government licence/fee; used by operators	Free to use (ISM bands), no licence
Less interference, guaranteed QoS, wide coverage	Interference-prone, no QoS guarantee
Example: NB-IoT / LTE-M (cellular)	**Example:** Wi-Fi, BLE, Zigbee (2.4 GHz), LoRa (868/915 MHz)

বাংলা: Licensed ব্যান্ডে সরকারি অনুমতি/ফি লাগে, হস্তক্ষেপ কম। Unlicensed ব্যান্ড (ISM) ফ্রি, কিন্তু ইন্টারফেরেন্স বেশি।

vii. What is edge computing in IoT? How does it differ from traditional cloud-based data processing?

Edge computing processes data near its source (on the device, gateway or local edge server) instead of sending everything to a distant cloud.

Edge computing	Traditional cloud
Processing near the device	Processing in a central data centre
Low latency, real-time response	Higher latency
Less bandwidth used	Large data uploaded
Works with poor connectivity, more private	Needs reliable Internet

বাংলা: এজ কম্পিউটিংয়ে ডেটা উৎসের কাছেই প্রসেস হয়, তাই দ্রুত ও কম ব্যান্ডউইথ লাগে। ক্লাউডে সব ডেটা দূরের সার্ভারে যায়, তাই দেরি হয়।

Question 2 (2 × 7 = 14)

[A] i. What are the six basic communication criteria used to describe IoT access technologies? Explain each briefly.

1. **Range:** the distance covered (short, medium, long).
2. **Frequency bands:** licensed or unlicensed spectrum used.
3. **Power consumption:** powered vs battery-operated (low power needed for years of life).
4. **Topology:** star, mesh, peer-to-peer.
5. **Constrained devices:** limits of the end devices' memory, CPU and power.
6. **Constrained-node networks:** low data rate, lossy links, limited bandwidth.

বাংলা: Range (দূরত্ব), Frequency band, Power consumption, Topology, Constrained devices, Constrained-node networks।

ii. Differentiate between short-range, medium-range, and long-range IoT access technologies with one example each.

Class	Range	Example
Short	up to ~100 m	Bluetooth Low Energy, Zigbee (802.15.4), NFC
Medium	~100 m to 1 km	Wi-Fi HaLow (802.11ah), 802.15.4g
Long	several km to tens of km	LoRaWAN, NB-IoT, Sigfox

iii. What are the three classes of constrained nodes defined by RFC 7228? State their RAM, flash, and IP stack capabilities.

Class	RAM	Flash	IP stack capability
Class 0	≪ 10 KB	≪ 100 KB	Cannot run an IP stack securely; needs a gateway/proxy
Class 1	~10 KB	~100 KB	Cannot run a full stack (HTTP/TLS) but can use a light stack (CoAP/UDP, 6LoWPAN)
Class 2	~50 KB	~250 KB	Can support a (lighter) full IP stack, similar protocols to normal nodes

বাংলা: Class 0 সবচেয়ে ছোট (গেটওয়ে লাগে), Class 1 হালকা স্ট্যাক (CoAP), Class 2 প্রায় পূর্ণ IP স্ট্যাক চালাতে পারে।

iv. List any five common security attacks in IoT. Explain any two of them briefly.

Five attacks: DoS/DDoS, Man-in-the-Middle, Eavesdropping (sniffing), Replay attack, Spoofing/Sybil, Botnet (e.g. Mirai), Physical tampering (any five).

- **DDoS:** thousands of compromised IoT devices (a botnet) flood a server/network with requests, so legitimate users cannot access it.
- **Man-in-the-Middle:** the attacker secretly sits between two communicating nodes, reading or altering messages (e.g. between a sensor and a gateway).

বাংলা: DDoS = অসংখ্য হ্যাক হওয়া ডিভাইস দিয়ে সার্ভারে ভিড় তৈরি করা। MITM = দুই পক্ষের মাঝখানে বসে ডেটা চুরি/পরিবর্তন করা।

v. What is a smart object in IoT? Describe its five essential characteristics as mentioned in the slides.

A smart object is a physical object with embedded computing, sensing and communication ability that can interact with its environment and other objects.

1. **Processing unit** (microcontroller) to process data.
2. **Sensor(s)/actuator(s)** to sense or act on the environment.
3. **Communication device** (wireless/wired) for networking.
4. **Power source** (battery, solar, energy harvesting).
5. **Unique identity** (address/ID) so it can be identified on the network.

বাংলা: স্মার্ট অবজেক্ট = প্রসেসর + সেন্সর/অ্যাকচুয়েটর + কমিউনিকেশন + পাওয়ার + ইউনিক আইডি। (নিজের স্লাইডের পাঁচটি পয়েন্ট মিলিয়ে নিও।)

vi. Draw and explain the pin diagram of Arduino UNO. Specify the functions of at least five pins.

```
            ARDUINO UNO (ATmega328P)
Digital: 0(RX) 1(TX) 2 ~3 4 ~5 ~6 7 | 8 ~9 ~10 ~11 12 13 GND AREF SDA SCL
Power  : IOREF  RESET  3.3V  5V  GND  GND  VIN
Analog : A0  A1  A2  A3  A4(SDA)  A5(SCL)
(~ = PWM pin)      USB port       DC jack (7-12 V)
```

- **D0 (RX), D1 (TX):** serial communication (UART).
- **D3, 5, 6, 9, 10, 11 (~):** PWM output, for motor speed and LED dimming.
- **A0-A5:** 10-bit analog inputs (0-5 V maps to 0-1023).
- **5V / 3.3V / GND:** supply for sensors and modules.

- **VIN:** external supply input (7-12 V).
- **RESET:** restarts the microcontroller.
- **A4 (SDA), A5 (SCL):** I^2C. **D10-D13:** SPI (SS, MOSI, MISO, SCK).

বাংলা: ১৪টি ডিজিটাল পিন (৬টি PWM), ৬টি অ্যানালগ ইনপুট, পাওয়ার পিন (5V, 3.3V, GND, VIN), RESET। RX/TX সিরিয়ালের জন্য, SDA/SCL I2C-এর জন্য।

vii. What are the two mandatory functions in an Arduino sketch? Explain their purposes with suitable examples.

- `void setup()` runs **once** at start/reset. It initialises pin modes (`pinMode`), serial (`Serial.begin`) and libraries.
- `void loop()` runs **repeatedly forever** after setup. It holds the main logic: reading sensors and controlling outputs.

```
void setup() { pinMode(13, OUTPUT); }
void loop()  { digitalWrite(13, HIGH); delay(1000);
               digitalWrite(13, LOW);  delay(1000); }
```

বাংলা: setup() একবার চলে (প্রাথমিক সেটিং), loop() বারবার চলে (মূল কাজ)। উদাহরণে LED ১ সেকেন্ড পরপর জ্বলে-নেভে।

Question 3

[A] (4 marks) i. Draw a chip select circuit to have the address of an I/O mapped I/O port as E7H.
ii. Draw a chip select circuit to have the address of a memory mapped I/O port as 7EE7H.

i. I/O-mapped port at E7H (8-bit address)

E7H = 1110 0111, so A7=1, A6=1, A5=1, A4=0, A3=0, A2=1, A1=1, A0=1.

```
A7 ----------------------\
A6 ----------------------\
A5 ----------------------\
A4 --[NOT]----------------\
A3 --[NOT]----------------| 8-input NAND |---- CS (active low)
A2 ------------------------/        |
A1 ----------------------/    IO/M = 1 (I/O cycle) also fed to gate
A0 ----------------------/
```

CS = A7 · A6 · A5 · A4' · A3' · A2 · A1 · A0 · (IO/M). Inverters are on A4 and A3 (the zero bits). IO/M (or IOR/IOW) is also fed to the gate so only an I/O cycle selects the port.

ii. Memory-mapped port at 7EE7H (16-bit address)

7EE7H = 0111 1110 1110 0111, so A15=0, A14-A9=1, A8=0, A7-A5=1, A4=0, A3=0, A2-A0=1.

```
Inverters on A15, A8, A4, A3 (zero bits); other lines direct.
All 16 address lines --> 16-input AND/NAND --> CS
Enabled also by MEMR / MEMW (IO/M' = 1, i.e. memory cycle)
```

CS = A15' · A14 · A13 · A12 · A11 · A10 · A9 · A8' · A7 · A6 · A5 · A4' · A3' · A2 · A1 · A0 · (memory cycle)

বাংলা: যে বিট 0 তাতে NOT গেট বসাতে হয়, যে বিট 1 সরাসরি যায়। সবগুলো AND/NAND গেটে দিলে ঠিক ওই ঠিকানায় CS সক্রিয় হয়। I/O-mapped-এ ৮টি লাইন, memory-mapped-এ ১৬টি লাইন লাগে এবং IO/M কন্ট্রোল সিগন্যাল ব্যবহার হয়।

[B] (5 marks) Assuming signed numbers, find the sign, carry, and overflow flags of: i. 09H + 17H ii. 71H + A9H iii. 6EH + 3AH iv. 7EH + 7BH

Sign = MSB of result; Carry = carry out of MSB; Overflow = (carry into MSB) XOR (carry out of MSB).

	Operation (8-bit)	Result	Sign	Carry	Overflow
i	0000 1001 + 0001 0111	20H (0010 0000)	0 (+)	0	0
ii	0111 0001 + 1010 1001	1AH (0001 1010)	0 (+)	1	0
iii	0110 1110 + 0011 1010	A8H (1010 1000)	1 (−)	0	**1**
iv	0111 1110 + 0111 1011	F9H (1111 1001)	1 (−)	0	**1**

বাংলা: (iii) ও (iv)-তে দুই ধনাত্মক সংখ্যা যোগ করে ঋণাত্মক ফল এসেছে (যেমন 110 + 58 = 168 > 127), তাই overflow = 1। (ii)-তে ধনাত্মক + ঋণাত্মক, তাই carry থাকলেও overflow নেই।

[C] (3 marks) In USB communication, a single transaction involves the transmission of up to three packets. Mention the names of these packets and describe the specific services each one provides during the communication process.

1. **Token packet** (sent by host): defines the type (IN / OUT / SETUP), the 7-bit device address and 4-bit endpoint number, with CRC5. It says which device/endpoint and the direction.
2. **Data packet** (by host or device): carries the payload (DATA0/DATA1 toggle for sequence check) with CRC16.
3. **Handshake packet** (by the receiver): reports the result: **ACK** (received OK), **NAK** (not ready), **STALL** (endpoint halted/error).

Flow: Host sends Token, then data is sent (OUT: host to device; IN: device to host), then the receiver replies with a Handshake. (Isochronous transfers have no handshake.)

বাংলা: Token = কে/কোথায়/কোন দিকে; Data = আসল ডেটা; Handshake = সফল হলো কি না (ACK/NAK/STALL)।

[D] (2 marks) What is meant by foldback in linear decoding?

In linear (partial) decoding, address lines are connected directly to chip selects and some address lines are left undecoded. So the same device responds to several different address ranges, and its address space "folds back" (repeats/mirrors) many times in the memory map. This wastes address space and makes the map non-unique, but it needs less hardware and is cheap.

বাংলা: লিনিয়ার ডিকোডিংয়ে কিছু অ্যাড্রেস লাইন ব্যবহার না করায় একই ডিভাইস একাধিক ঠিকানায় সাড়া দেয়। একে foldback (mirror/image) বলে।

Question 4

[A] (5 marks) Show the pin configuration of 8255 PPI and separately describe the operation performed by each unit.

The 8255A is a 40-pin programmable parallel interface with 24 I/O lines (three 8-bit ports).

Pin	Name	Pin	Name	Pin	Name	Pin	Name
1	PA3	11	PC6	21	PB3	31	D3
2	PA2	12	PC5	22	PB4	32	D2
3	PA1	13	PC4	23	PB5	33	D1
4	PA0	14	PC0	24	PB6	34	D0
5	RD'	15	PC1	25	PB7	35	RESET
6	CS'	16	PC2	26	Vcc	36	WR'
7	GND	17	PC3	27	D7	37	PA7
8	A1	18	PB0	28	D6	38	PA6
9	A0	19	PB1	29	D5	39	PA5
10	PC7	20	PB2	30	D4	40	PA4

Pin groups: D0-D7 (data bus), RD', WR', CS', A0, A1, RESET, PA0-7, PB0-7, PC0-7, Vcc, GND.

Operation of each unit

- **Data bus buffer:** 8-bit tri-state bidirectional buffer connecting the 8255 to the CPU data bus.
- **Read/Write control logic:** uses RD', WR', CS', A0, A1, RESET to select a port/control register and the direction of transfer (A1A0 = 00 Port A, 01 Port B, 10 Port C, 11 Control register).
- **Group A control:** controls Port A and upper Port C (PC7-PC4).
- **Group B control:** controls Port B and lower Port C (PC3-PC0).
- **Ports A, B, C:** data I/O ports programmed as input or output. Port C can be split into two 4-bit halves, used for handshake/control signals or bit set/reset.
- **Modes:** Mode 0 (simple I/O), Mode 1 (strobed I/O), Mode 2 (bidirectional, Port A only), BSR mode (bit set/reset on Port C).

> **বাংলা:** 8255-এ ৩টি পোর্ট (A, B, C), data buffer, read/write control এবং Group A/B কন্ট্রোল ইউনিট আছে। A1A0 দিয়ে পোর্ট বা কন্ট্রোল রেজিস্টার বাছাই হয়।

[B] (4 marks) When you press a key on your computer, you are activating a switch. There are many different ways of making these switches. Describe any four of them.

When a key is pressed, a switch contact closes (or changes) and the keyboard controller detects this as a key code. Four ways of making switches:

1. **Mechanical (contact) switch:** metal contacts touch when pressed. Cheap, but contact bounce occurs (needs debouncing).
2. **Membrane switch:** two conductive layers separated by a spacer touch when pressed. Thin, spill-proof.
3. **Capacitive switch:** a key press changes the capacitance between plates and the circuit senses it. No physical contact, long life.
4. **Hall-effect switch:** a magnet moves near a Hall sensor and the voltage change gives the key signal. Contactless, very reliable.

(Also: reed switch, conductive-rubber switch.)

বাংলা: কী চাপলে সুইচ কন্ট্যাক্ট বদলায় এবং কন্ট্রোলার সেটি ধরে। সুইচ তৈরির ধরন: Mechanical, Membrane, Capacitive, Hall-effect।

[C] (3 marks) i. How many address and data lines are required for a 1M × 16 memory chip?
ii. What is the size of a decoder with one chip enable (CE) to obtain a 64K × 32 memory from 4K × 8 chips? Where are the inputs and outputs of the decoder connected?

i. 1M = 2^{20} locations, so **20 address lines** (A0-A19). Word width 16 bits, so **16 data lines** (D0-D15).

ii.

- Chips needed = (64K / 4K) × (32 / 8) = 16 × 4 = **64 chips**.
- 64K needs 16 address lines (A0-A15). Each 4K chip needs 12 (A0-A11), so the remaining **4 lines (A12-A15)** select one of 16 rows.
- So the decoder is a **4-to-16 decoder** with one enable (CE).
- **Inputs:** A15-A12 go to the decoder inputs; CE is driven by the memory-request control signal.
- **Outputs:** 16 outputs, each goes to the chip-select of one row of 4 chips (4 × 8 = 32 data bits). A0-A11 are connected in parallel to all chips; the four chips in a row supply different bytes of D0-D31.

বাংলা: 1M = 2^{20}, তাই ২০ অ্যাড্রেস লাইন ও ১৬ ডেটা লাইন। 64K×32 বানাতে ৬৪টি 4K×8 চিপ লাগে; A12-A15 কে 4-to-16 ডিকোডারের ইনপুটে দিতে হয়, আর ১৬টি আউটপুট প্রতিটি ৪টি চিপের সারির CS-এ যায়।

[D] (2 marks) Write down the advantages and disadvantages of PDP over CRT and LCD.

Advantages	Disadvantages
Thin, flat and lighter than CRT; large screen sizes; no geometric distortion or magnetic interference; wide viewing angle; high contrast/true blacks and fast response (better than LCD)	High power consumption and heat; screen burn-in (image retention); expensive; heavier than LCD; poor at high altitude; hard to make in small sizes; shorter life

বাংলা: সুবিধা: পাতলা, বড় স্ক্রিন, বিস্তৃত ভিউ অ্যাঙ্গেল, ভালো কনট্রাস্ট। অসুবিধা: বেশি বিদ্যুৎ খরচ, গরম হওয়া, burn-in, দাম বেশি।

Question 5

[A] (5 marks)
i. If we want to cut a 512 × 512 sub-image out from the centre of an 800 × 600 image, what are the coordinates of the pixel in the large image that is at the lower left corner of the small image?
ii. Find the CMY coordinates of a colour at (0.2, 1, 0.5) in the RGB space.
iii. If we use direct coding of RGB values with 2 bits per primary colour, how many possible colours do we have for each pixel?
iv. What is the pitch of a colour CRT?
v. Why do many colour printers use black pigment?

i. Centre of large image = (400, 300). Half of the sub-image = 256.
x from 400 − 256 = **144** to 656; y from 300 − 256 = **44** to 556.
Lower-left corner = (144, 44) with the origin at bottom-left. (If the origin is top-left with y downward, it is (144, 555).)

ii. CMY = 1 − RGB, so C = 1 − 0.2 = 0.8, M = 1 − 1 = 0, Y = 1 − 0.5 = 0.5. Answer: **(0.8, 0, 0.5)**.

iii. 2 bits per primary gives $2^2 = 4$ levels each, so 4 × 4 × 4 = **64 colours** (6 bits per pixel).

iv. The **pitch** of a colour CRT is the centre-to-centre distance between two neighbouring phosphor dots (triads) of the same colour (typically 0.25-0.31 mm). Smaller pitch gives a sharper, higher-resolution image.

v. Mixing C+M+Y should in theory give black, but real inks give a muddy dark brown. Black ink gives deeper, sharper text, is cheaper than using three colour inks, uses less ink so paper dries faster, and gives better quality (CMYK).

বাংলা: (i) কেন্দ্র (400,300) থেকে ২৫৬ বাদ দিলে নিচের-বাম কোণ (144,44)। (ii) CMY = 1 − RGB = (0.8, 0, 0.5)। (iii) 4^3 = ৬৪ রং। (iv) একই রঙের দুই ফসফর ডটের কেন্দ্র থেকে কেন্দ্রের দূরত্ব। (v) CMY মিশিয়ে আসল কালো হয় না, তাই আলাদা কালো কালি লাগে।

[B] (4 marks) Illustrate the key components of an inkjet printer and describe the function of each component performs.

- **Print head and nozzles:** sprays tiny ink droplets onto paper (thermal: heating element makes a vapour bubble; or piezoelectric: crystal deforms).
- **Ink cartridges:** hold CMYK ink and supply the print head.
- **Carriage, stepper motor and belt:** moves the print head horizontally across the paper.
- **Paper feed mechanism (rollers + motor):** advances the paper line by line.
- **Control circuit (controller board/memory):** receives data from the computer, converts it to nozzle firing signals, and controls the motors.
- **Power supply and interface (USB/Wi-Fi):** supplies power and connects to the computer.

বাংলা: প্রিন্ট হেড (কালি ছিটায়), কার্টিজ (কালি রাখে), ক্যারেজ মোটর (হেড সরায়), পেপার ফিড রোলার (কাগজ টানে), কন্ট্রোলার বোর্ড (সিগন্যাল নিয়ন্ত্রণ)।

[C] (3 marks) Describe the operational modes of 8279.

The 8279 is a programmable keyboard/display interface. Modes are set by writing a command word to its control register.

- **Keyboard modes:** (1) scanned keyboard with 2-key lockout; (2) scanned keyboard with N-key lockout; (3) scanned keyboard with N-key rollover; (4) scanned sensor matrix; (5) strobed input (data latched on strobe).
- **Display modes:** 8- or 16-character display; *left entry* (typewriter style) or *right entry* (calculator style).

বাংলা: কীবোর্ড মোড: 2-key lockout, N-key lockout, N-key rollover, sensor matrix, strobed input। ডিসপ্লে মোড: ৮/১৬ ক্যারেক্টার, left/right entry।

[D] (2 marks) Give the pinout of USB Type C 3.0.

24 pins in two rows (A and B, 12 pins each):

Pin	Row A	Row B
1	GND	GND
2	TX1+	TX2+
3	TX1−	TX2−
4	VBUS	VBUS
5	CC1	CC2 (VCONN)
6	D+	D+
7	D−	D−
8	SBU1	SBU2
9	VBUS	VBUS
10	RX2−	RX1−
11	RX2+	RX1+
12	GND	GND

বাংলা: Type-C তে ২৪টি পিন: GND, VBUS (পাওয়ার), TX/RX (হাই-স্পিড ডেটা), D+/D− (USB 2.0), CC (কানেকশন/ওরিয়েন্টেশন), SBU (সাইডব্যান্ড)।

Question 6

[A] (5 marks) Differentiate between: i. Standard and memory-mapped programmed I/O ii. Maskable and non-maskable interrupts iii. Internal and external interrupts iv. Block transfer and interleaved DMA v. Polled and daisy chain interrupts

	First	Second
i	**Standard (isolated) I/O:** separate 256-address I/O space, uses IN/OUT and the IO/M signal; does not use memory space	**Memory-mapped I/O:** ports have memory addresses; any memory instruction (MOV, LDA) can be used; uses up memory space
ii	**Maskable:** can be enabled/disabled by software (e.g. RST 7.5, INTR)	**Non-maskable:** cannot be disabled, highest priority (e.g. TRAP); used for power failure, critical errors
iii	**Internal:** caused inside the CPU (divide by zero, overflow, software INT n)	**External:** caused by external hardware through pins (INTR, NMI, I/O device)
iv	**Block (burst) DMA:** the DMAC takes the bus and transfers a whole block; CPU is halted until done	**Interleaved DMA:** DMA transfers one word when the CPU is not using the bus; CPU not stopped, but transfer is slower
v	**Polled:** CPU checks each device's status in software order to find the interrupter; simple but slow	**Daisy chain:** devices connected in series with an acknowledge line; nearest device gets priority; hardware identifies the source; faster

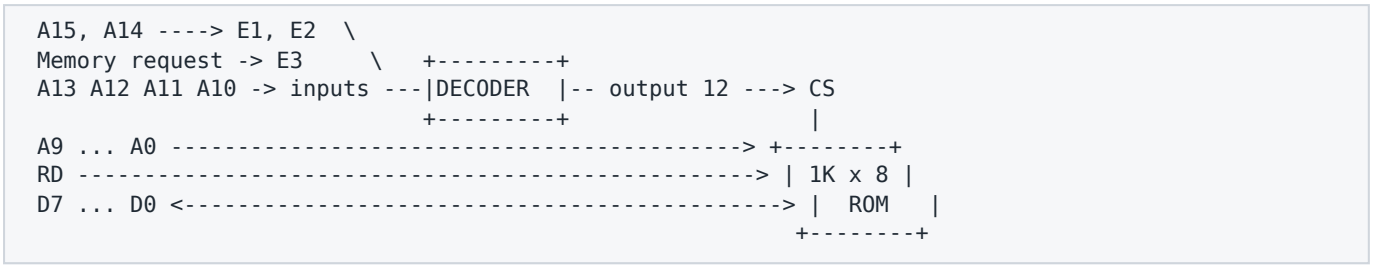
বাংলা: Memory-mapped মেমোরি ঠিকানা ব্যবহার করে; NMI বন্ধ করা যায় না; Internal interrupt CPU-র ভেতরে তৈরি হয়; Block DMA-তে CPU থেমে যায়; Daisy chain-এ হার্ডওয়্যারে প্রায়োরিটি ঠিক হয়, Polled-এ সফটওয়্যারে একে একে চেক করা হয়।

[B] (5 marks) Use the following chips (Figure 6.1: a decoder with inputs 13-10, enables E1/E2/E3, and a 1K × 8 ROM with CS, RD, D0-D7) to design a microcomputer memory to include the map 3000H through 33FFH. Draw a neat logic diagram.

- Range size = 33FFH − 3000H + 1 = 400H = **1 KB**, so one **1K × 8 ROM** is needed (10 address lines A0-A9).
- 3000H = 0011 0000 0000 0000, so for the whole range A15=0, A14=0, A13=1, A12=1, A11=0, A10=0 (A9-A0 vary from 000H to 3FFH).

Connections

- **A0-A9** from the microprocessor connect directly to the ROM address pins A0-A9.
- **A10, A11, A12, A13** connect to the decoder inputs (lines 10, 11, 12, 13). A13A12A11A10 = 1100 = 12, so output **12** goes active.
- **A15, A14** are used to enable the decoder (enabled only when both are 0, e.g. through an OR/NOR gate to E1/E2); the remaining enable (E3) is driven by the memory-request (MEMR) signal or tied to its enabled level.
- Decoder **output 12** goes to the **CS** of the ROM.
- **D0-D7** of the ROM connect to the microprocessor data bus; **RD** from the microprocessor connects to the ROM RD/OE.



Draw the figure neatly in the exam using the actual pin labels of Figure 6.1. If the decoder in your paper has different pin numbering, the same logic applies: A13A12A11A10 = 1100 selects the ROM.

বাংলা: 3000H-33FFH = ১ KB, তাই একটি 1K×8 ROM। A0-A9 সরাসরি ROM-এ যায়, A10-A13 ডিকোডারের ইনপুটে (মান 1100 = ১২ নম্বর আউটপুট সক্রিয়) এবং A15, A14 (দুটোই 0) ডিকোডার এনেবল করে। ১২ নম্বর আউটপুট ROM-এর CS-এ যায়।

[C] (2 marks) Write short notes on: i. Protocol Tunneling ii. USB4 Fabric

i. Protocol tunneling: In USB4, different protocols (PCIe, DisplayPort, USB 3.x) are encapsulated into USB4 packets and carried over the single USB4 link. The tunnelled protocols share the 40 Gbps bandwidth dynamically, so one cable can carry display, storage and data at once, without changing the original protocols.

ii. USB4 fabric: The connection topology of a host router, device routers and hubs linked by USB4 links (based on Thunderbolt 3), which route tunnelled packets between devices. It supports a tree topology, 2 lanes × 20 Gbps = 40 Gbps, dynamic bandwidth allocation, and backward compatibility with USB 3.2/USB 2.0 and Thunderbolt 3.

বাংলা: Protocol tunneling = অন্য প্রোটোকল (PCIe, DisplayPort) USB4 প্যাকেটের ভেতরে ঢুকিয়ে এক তারে পাঠানো। USB4 fabric = router ও link দিয়ে তৈরি নেটওয়ার্ক কাঠামো (৪০ Gbps)।

[D] (2 marks) What are the functions of CCD and CIS inside a scanner?

- **CCD (Charge-Coupled Device):** a row of light-sensitive cells. Light reflected from the document passes via mirrors and a lens onto the sensor and is converted to electrical charge, which is shifted out and digitised (ADC). Gives high quality and good depth of field, but needs a lens/mirror system (bulky, higher power).
- **CIS (Contact Image Sensor):** RGB LEDs illuminate the document and a rod-lens array focuses light directly onto the sensor strip very close to the paper. Compact, light, cheap and low power, but lower depth of field and slightly lower quality.

> **বাংলা:** দুটোই ডকুমেন্টের আলো বৈদ্যুতিক সিগন্যালে বদলায়। CCD আয়না ও লেন্স ব্যবহার করে (মান ভালো, আকারে বড়), CIS কাগজের খুব কাছে সেন্সর ও LED রাখে (ছোট, সস্তা, মান কিছুটা কম)।